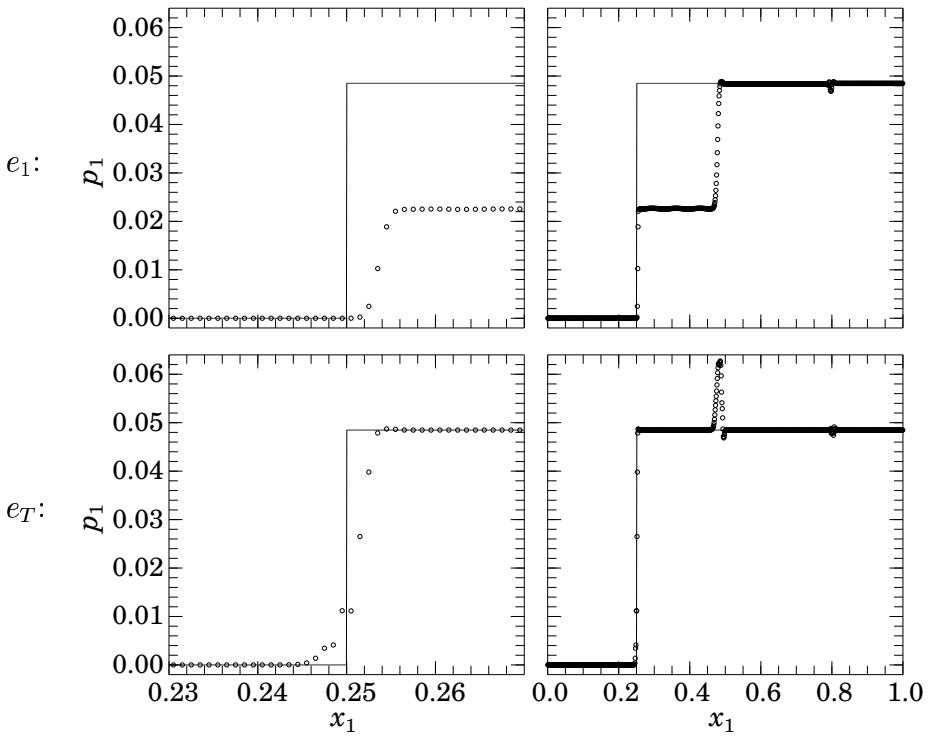
ZEUS-3D 1-D Gallery #15: Convergence tests



This is from Fig. 5 of Falle, 2002 (ApJ, 577, L123), depicting the pressure (p_1) from the MHD shock tube test with the left state $(\rho, v_1, v_2, v_3, B_2, B_3, p_1) = (1, 1.5, 0, 0, 1, 0, 10^{-6})$ and the right state $(1.6111, 0.9310, 0.04104, 0, 1.6156, 0, 0.04847)$ with $B_1 = 0.1$, $\gamma = 5/3$ at time $t = 0.25$. At $t = 0$, the discontinuity is at $x_1 = 0.25$. These levels satisfy the Rankin-Hugoniot jump conditions for a stationary (in the grid frame) fast shock and thus, ideally, the numerics should just "hold onto" these initial conditions.

Open circles are the `dzeus35` solution using 1,000 zones, `CMoC`, the internal/total energy equation on the top/bottom row (labelled e_1/e_T), and second-order interpolation. `dzeus35` parameters controlling the time step and artificial viscosity are: `courno=0.5`, `qcon=2.0`, and `qlin=0.2`. Left panels depict the region immediately around the shock; right panels show the entire grid. Lines are the results from the non-linear Riemann solver described in Ryu & Jones (1995, ApJ, 442, 228).

Falle's (2002) assertion that *ZEUS* does not always converge to the conservative solution of the Riemann problem is exemplified by the top panels. In this extreme case (pressure jump $\sim 5 \times 10^4$, $\beta_L = 2 \times 10^{-6}$, $\beta_R = 0.037$), the non-conservative internal energy equation inserts an additional level to the right of the shock whose width grows with the slow magnetosonic speed. This should be viewed not so much as a failure of the code, but rather an inappropriate use: there can be no surprise that a non-conservative system of difference equations does not give the solution to a conservative set of differential equations.

As the bottom panels show, `dzeus35` in its conservative mode (Clarke, 1996) is able to hold onto the shock satisfactorily without inserting any additional levels. Unlike the purely upwinded scheme used by Falle (2002), a one-zone description of a standing shock is *not* a steady-state solution to the difference equations solved by `dzeus35` and thus the imposed initial conditions are not maintained. Instead, `dzeus35` launches two left- and right-moving transient waves (one each at the slow and fast magnetosonic speeds) from the one-zone shock leaving behind a several-zone description of the standing shock with the correct pre- and post-shock levels intact. As this *is* a steady-state solution to the conservative difference equations in `dzeus35`, this is the solution the scheme holds onto. Note that because of the *very* slow sound speed in $x < 0.25$, the left-moving transients are still in the process of leaving the shock at $t = 0.25$.